

- Q.25 Give the principle of pacemaker. (CO5)
- Q.26 How the clinical laboratory equipment differs from diagnostic equipment? (CO1)
- Q.27 Calorimeter is used for _____. (CO3)
- Q.28 Explain briefly the ultrasonic imaging system. (CO5)
- Q.29 Discuss the role of 'Bucky Grid' in X-Ray machine. (CO5)
- Q.30 Discuss temperature sensor. (CO3)
- Q.31 Give basic idea about contact impedance. (CO2)
- Q.32 Draw and explain the working of PCG. (CO5)
- Q.33 Classify the bio electrodes on the basis of application. (CO2)
- Q.34 Give the principle of defibrillator. (CO5)
- Q.35 What is the use of nebulizer? (CO5)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Draw and explain the block diagram of ECG machine. (CO4)
- Q.37 Explain any one therapeutic equipment in detail with the help of diagram. (CO1)
- Q.38 Write a short note on : (CO5,6)
- Heart rate measurement
 - Safety standards for medical instruments.

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Roll No.

4th Sem / Eltx Subject:- Medical Electronics

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Active transducers work on the principle of _____ (CO3)
- energy conversion
 - mass conversion
 - energy alteration
 - volume conversion
- Q.2 What is the frequency range of ECG machine?(CO4)
- 5 to 20 Hz
 - 0.05 to 120 Hz
 - 120-200 Hz
 - 200-500 Hz
- Q.3 Principle behind strain gauge is _____ (CO3)
- Variable resistance
 - variable inductance
 - variable capacitance
 - variable contact area
- Q.4 Spirometer is used for _____ measurement. (CO3)
- Respiration
 - Heart rate
 - Pulse rate
 - None of above
- Q.5 Which of the following instrument is used for recording the electrical activity of the brain? (CO1)
- ECG
 - EMG
 - PCG
 - EEG

Q.6 Which instrument is used for clinical detection of heart sounds? (CO1)

- a) Stethoscope b) Endoscope
- c) Anoscope d) Proctoscope

Q.7 Gauge factor is defined as _____ (CO3)

- a) (incremental change in resistance due to stress / resistance of an unstretched wire)* (unstretched length of wire / incremental change in length)
- b) (incremental change in resistance due to strain/ resistance of an unstretched wire)* (unstretched length of wire / incremental change in length)
- c) (incremental change in resistance due to stress/ resistance of an unstretched wire)*(incremental change in length / unstretched length of wire)
- d) (resistance of an unstretched wire/incremental change in resistance due to stress)* (unstretched length of wire / incremental change in length)

Q.8 Which of the following is not a piezo-electric material? (CO3)

- a) quartz b) Rochelle salt
- c) Aluminium d) barium titanate

Q.9 LVDT stands for _____ (CO3)

- a) Linear Virtual Double Transformer
- b) Linear Virtual Differential Transducer
- c) Linear Variable Differential Transducer
- d) Linear Variable Differential Transformer

Q.10 The smallest change in measurand that will result in a measurable change in the transducer output is called _____ (CO3)

- a) offset b) linearity
- c) resolution d) threshold

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 MRI stand for _____. (CO5)

Q.12 The device used to measure the absorbance of specific solution or emitted heat is known as _____. (CO3)

Q.13 Name any two therapeutic equipment? (CO1)

Q.14 What is a gross shock? (CO6)

Q.15 Define contact impedance. (CO2)

Q.16 Define Electrodes. (CO2)

Q.17 Define ECG. (CO4)

Q.18 Define Respiration Rate. (CO3)

Q.19 Define Therapeutic Equipment. (CO1)

Q.20 EEG stand for _____. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write any five precautions to minimize electric shock hazards from biomedical equipment. (CO6)

Q.22 Explain Heart rate measurement system. (CO5)

Q.23 Write a short note on respiration sensor. (CO3)

Q.24 What is Micro current shock. (CO6)